electronics," explains Karan.

He adds, "There are various opportunities at each juncture of the electronics supply chain as well, and these can be very big!"

The FICCI Circular Economy Report, 2017 outlined that the business opportunity for extracting gold from e-waste is to the tune of $0.7 to $1 billion. It is worth noting that one tonne of ore has an extractable reserve of about 1.4gm gold while a tonne of mobile phone PCBs can yield about 1.5kg. Global E-waste Monitor 2020 estimates that in 2019 as only 17.4 percent of e-waste was collected and recycled, it resulted in a loss of nearly $47 billion in recoverable materials including gold, silver, copper, platinum, and other high-value materials.

"The government has covered 21 categories under the electric waste rules. As of now, there are categories beyond the ambit of ones that have already been covered. The NITI Ayog paper has recommended covering all the left categories under the e-waste rules," says Pranshu. "Plastics and regular metals form the main part of e-waste. Then some categories also contain precious metals like gold, silver," he adds.

Collection and segregation

The very first opportunity offered by the world of e-waste starts with an efficient collection process. It is mandated by law that sellers of electronic goods like smartphones, smart TVs, CFL bulbs, and most other consumer electronics goods, need to collect some percentage of electronics they sell annually. This means that for every 100 smartphones that a brand sells, it is the brand's responsibility to collect at least some at end of lifespan of the product. The average lifespan of various consumer electronics categories differs widely.

"Being able to collect these products at the end of the lifespan is a big problem for a lot of brands. If a company, for that matter a startup, can come up with a plan to effectively collect such products, it will be of great help to the original producers. Collaborating with brands and being able to plan and implement collection of e-waste is a very big business opportunity itself," Karan explains.

"Lithium-ion batteries, though a small part of e-waste, is the talk of the town right now. Apart from being a very important subject right now, the recycling opportunity in this domain is huge enough to create economies of scale. The best part about e-waste is everything being interconnected," adds Karan.

Some of the best ways to collect e-waste include partnering with municipalities, societies, and waste traders. While the former two can help one source e-waste directly, the latter (may be a more efficient one) can help source e-waste in a faster manner as waste traders usually have their own channel partners who collect waste and sell it to them.

"Tying up with traders can help in the long term as you can guide these traders on what to collect and how to collect. These traders, in turn, can teach their channel partners the same. The scale of channel partners such traders have can be highlighted from the number of scrap buyers (*kabbadi walas*) roaming on the streets asking for scrap in exchange of money," says Karan.

The next problem, which is also a big business opportunity, is segregation of the waste. Apart from brands who are trying to meet their e-waste collection targets, one cannot just sell entire smartphones, laptops, or other electronic good just like that. There are also dealers who buy outdated electronic goods in bulk, but the bigger opportunity lies in being able to segregate this waste and sell it in batches accordingly.

Pranshu says, "Setting up a system that enables effective e-waste collection management is the biggest opportunity of them all. Neither do we have evolved collection systems nor do we have the best-in-class recycling systems for multiple categories in India." EFY

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